WEEK 4 • MODULE 9

// DEPLOYMENT READINESS & APP DESIGN

DEPLOYMENT READINESS & STREAMLIT APP DESIGN.

FINAL PIPELINE AUDIT & CUSTOMER SEGMENTATION DASHBOARD PLAN

Reviewing the dataset, engineered features, saved model, and segment labels produced across the project, then planning the content, navigation, and technology stack for the Streamlit dashboard.

AI LAB 99

DATASET	MODEL	SCALER	SEGMENTS
2,038 rows × 42 cols	K-Means (kmeans_model.pkl)	standard_scaler.pkl — found	2 — Premium Loyal / Digital-First

TASKS 1–7 — PIPELINE AUDIT

- ✓ **Dataset finalized:** 2,038 × 42, matches Module 7/8 output exactly
- ✓ **13/13 engineered features present** and confirmed float64 (clustering-ready)
- ✓ **Model finalized:** K-Means, saved as kmeans_model.pkl
- ✓ **Scaler located:** standard_scaler.pkl saved alongside the model
- ✓ **Segments confirmed:** Cluster 0 / Cluster 1 map to 2 business names
- ✓ **Business segments:** Premium Loyal Customers, Digital-First Buyers

⚠ NAMING DRIFT ACROSS MODULES

Cluster 1 has had **three different names**: "Digital-First Buyers" (Module 7) → "Budget Buyers" (Module 8) → "Digital-First Buyers" again (Module 9, since it reloads the original Module 7 label). Pick one canonical name before the dashboard ships.

TASK 10 — TECHNOLOGY

Selected: Streamlit — Python-native, integrates with the existing pandas/scikit-learn pipeline, loads the model via Joblib.

Python · Streamlit · Pandas · Plotly · Joblib · scikit-learn

TASK 8 — DASHBOARD CONTENT 12 items

- ✓ Total number of customers
- ✓ Total number of segments
- ✓ Customer distribution by segment
- ✓ Income comparison ⚠
- ✓ Total spending comparison ⚠
- ✓ Product preferences
- ✓ Shopping channel preferences
- ✓ Customer activity levels
- ✓ Campaign response ⚠
- ✓ Customer persona
- ✓ Customer segment prediction
- ✓ Marketing recommendations

⚠ 3 PANELS AT RISK

Income / Spending comparison: will show meaningless z-scores (e.g. "−1.9") unless wrapped in `scaler.inverse_transform()`. **Campaign response:** upstream data is entirely zero (Module 3 outlier-capping bug) — will show nothing meaningful until fixed.

TASK 9 — NAVIGATION 8 pages

1. Home
2. Dataset Overview
3. Customer Segments
4. Business Dashboard
5. Customer Personas
6. **Predict Customer Segment** — needs full preprocessing chain applied to new input
7. **Marketing Recommendations** — feeds from Module 8's currently-empty strategy matrix

DEPLOYMENT READINESS

NEARLY READY

Dataset, features, model, and scaler are all in place. Three fixes remain before the dashboard is safe to ship to business users.

6/6 AUDIT ITEMS PASS

3 RISKS TO FIX

PRIORITY FIX LIST

1. **Inverse-transform** Income/spending columns before displaying anywhere in the dashboard
2. **Fix Module 8's join bug** before wiring the Marketing Recommendations page
3. **Settle on one segment name** per cluster — "Digital-First Buyers" recommended

PREDICTION PAGE REQUIREMENT

New customer input must pass through the **same chain** used in training: missing-value handling → feature engineering → encoding → StandardScaler — before calling `kmeans_model.predict()`. Reuse the Module 5 `preprocessing_pipeline()` function to guarantee this.